PATENTFLOW

Patent Filing Application

Complete Blueprint & Algorithm

A Step-by-Step Technical Guide for Building a
USPTO-Compliant Self-Service Patent Filing Platform

USA First
USPTO Compliant

All Patent Types
Utility, Design, Plant

AI-Assisted
Smart Drafting Engine

Up to 80% Savings
vs. Law Firm Cost

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1. EXECUTIVE SUMMARY & VISION

PatentFlow is a web-based SaaS platform that dramatically reduces or eliminates the need for expensive patent attorneys. Law firms charge **$8,000–$25,000+** for utility patents and **$2,000–$5,000** for design patents. PatentFlow targets **70–85% savings** by guiding inventors through every step — from disclosure through USPTO submission — using AI-assisted drafting, automated prior art search, and direct USPTO Patent Center integration.

Reference Patents Used: This blueprint was built using actual filed patents: US Design Patent D880,093 S ('Trocar Port Cleaning Device', Hoftman, filed Jul 28 2017, granted Mar 31 2020) and US Design Patent D723,689 S ('Surgical Scissors', George, 2015). These demonstrate the exact USPTO structure PatentFlow must generate.

Core Value Propositions

Value Proposition	Description
Cost Reduction	From $10,000+ down to under $2,000 — automate drafting, search, and filing
Guided Wizard	Step-by-step workflow for inventors with no legal background
AI-Assisted Drafting	LLMs generate specification, claims, and abstract from plain-English description
Automated Prior Art Search	USPTO PPUBS, Google Patents (87M+ patents), EPO Espacenet
Compliance Enforcement	Real-time validation: 37 CFR Part 1, MPEP, USPTO form requirements
Direct E-Filing	Submit directly to USPTO Patent Center via secure API

2. PATENT TYPES OVERVIEW (USPTO)

2.1 Utility Patents

Protect **how an invention works** — any new and useful process, machine, article of manufacture, or composition of matter. Most common and valuable patent type. Term: **20 years from nonprovisional filing date**, subject to maintenance fees at 3.5, 7.5, and 11.5 years.

- Requires: Specification (all sections), Claims (independent + dependent), Abstract, Drawings
- Filing fees (2025/2026): ~$1,850 total (micro); ~$3,700 (small); ~$7,400+ (large) — including search, examination, and issue fees
- Examination timeline: Average 22–30 months to first office action

2.2 Design Patents

Protect the **ornamental (visual) appearance** of a functional article — its shape, configuration, or surface ornamentation. Contain only **one claim** and rely primarily on drawings. Reference patents (D880,093 S and D723,689 S) are both design patents. **Note:** The Rocket Docket expedited program was eliminated August 14, 2025.

- Term: 15 years from grant date (for apps filed after May 13, 2015)
- Single claim: 'The ornamental design for [article], as shown and described.'
- Requires: all standard drawing views — front, back, left, right, top, bottom, perspective
- No maintenance fees required
- Fees (2025/2026): ~$1,300 total (large); ~$520 (small); ~$260 (micro)

2.3 Provisional vs. Nonprovisional

Feature	Provisional	Nonprovisional
Examined?	No — auto-abandoned after 12 months	Yes — full USPTO examination
Cost	$64–$320 (filing fee only)	$400–$3,500+ (filing + search + exam)
Term	12 months, then abandoned	20 years from filing (utility)
Claims required?	No	Yes — mandatory
Purpose	Establishes priority date; buys 12 months	Actual patent application
Patent Pending?	Yes, immediately	Yes, during examination

3. US PATENT DOCUMENT STRUCTURE

3.1 Utility Patent Application — Required Sections

#	Step	Description / Logic
1	**Title**	2–7 words, descriptive, max 500 chars. No articles ('a', 'an', 'the'). Example: 'Trocar Port Cleaning Device'.
2	**Cross-References**	Required if claiming priority from provisional, CIP, or continuation. Cite application number(s), filing date(s), relationship.
3	**Background**	Field of invention + problems with existing solutions. 1–3 paragraphs. Avoid admissions of prior art that could limit claim scope.
4	**Brief Summary**	Concise statement of the invention. Must parallel broadest independent claim without using claims language.
5	**Figure Descriptions**	One line per figure: 'FIG. 1 is a perspective view of the device of the present invention; FIG. 2 is a front elevational view thereof.'
6	**Detailed Description**	Complete enabling description referencing every numbered element in every figure. POSITA must be able to reproduce invention. 10–30+ pages for complex inventions.
7	**Claims**	Legal scope of protection. Up to 20 total (3 independent) in base fee. Each claim is one sentence. Must be supported by specification.
8	**Abstract**	Max 150 words. Technical summary for public and examiners. Mention key novel features. No claim language.
9	**Seq/Code Appendix**	Biotech (ST.26 XML) or software (large code). Submitted separately. Not required for most mechanical/design patents.

3.2 Design Patent Sections (Reference: D880,093 S and D723,689 S)

#	Step	Description / Logic
1	**Title/Subject**	Article name: e.g., 'TROCAR PORT CLEANING DEVICE'.
2	**Applicant/Inventor Info**	Inventor legal name, city, state, country. Assignee. Application number. Filing date. LOC, USPC, CPC classification codes. Field of Classification Search.
3	**References Cited**	All prior U.S. patents, foreign patents, and publications — examiner-added and applicant-disclosed. Patent number, date, inventor, classification code.
4	**Single Claim**	Exactly one claim: 'The ornamental design for [ARTICLE], as shown and described.'
5	**Description of Views**	'FIG. 1 is a perspective view; FIG. 2 is a front view; FIG. 3 is a back view; FIG. 4 is a right side view; FIG. 5 is a left side view; FIG. 6 is a top view; FIG. 7 is a bottom view thereof.'
6	**Drawing Sheets**	B&W; line drawings per 37 CFR § 1.84. Correct margins, line weight, sheet labels (Sheet X of Y), inventor name, figure numbers. Typically 4–7 sheets.

4. MASTER ALGORITHM — 10 PHASES

PatentFlow consists of 10 sequential phases, each a discrete module. Users progress forward but may return to revise earlier sections. AI assistance is available throughout every phase.

PHASE 1: USER ONBOARDING & IDENTITY VERIFICATION

#	Step	Description / Logic
1.1	**Registration**	Full legal name, email, phone, country. Email OTP verification. Generate user UUID. Encrypt credentials.
1.2	**KYC Verify**	SSN last 4 OR passport (foreign). Validate via KYC API (Persona/Jumio). Assign role: inventor / applicant / assignee.
1.3	**Entity Size**	CRITICAL for fee calculation. Questionnaire: Is applicant a person, small business (<500 employees), or large corporation? Has any rights been assigned to a large entity? Result: Large / Small Entity (50% fee discount) / Micro Entity (80% discount, income threshold applies). Store entity_type.
1.4	**Inventor Profile**	Legal name, citizenship, residence address (required for oath/declaration). Multiple co-inventors allowed — each must sign declaration. Store co-inventor relationships.
1.5	**Attorney Question**	Pro se (self-filing) or registered patent attorney/agent? If attorney: collect USPTO reg number, validate via roster API. If pro se: show disclaimer about complexity.
1.6	**Dashboard Init**	Create user workspace. Initialize application record (status=DRAFT). Display onboarding checklist. Show estimated total cost based on entity type.

PHASE 2: INVENTION DISCLOSURE INTAKE

#	Step	Description / Logic
2.1	**Invention Title**	'What do you call your invention? (2–7 words).' Validate length, no articles. AI suggests 3 alternatives.
2.2	**Plain Description**	Open text: 'Describe your invention — what it is, what it does, why it's better than existing solutions.' Min 100 words. AI parses key technical elements.
2.3	**Problem Being Solved**	'What problem does your invention solve? What are the shortcomings of current solutions?' Maps to Background section.
2.4	**How It Works**	Step-by-step operation. Physical: materials, components. Software: process flow. Chemical: composition and method. Maps to Detailed Description.
2.5	**Key Novel Features**	List 3–10 unique features. Each becomes potential independent or dependent claim element. AI suggests patent claim language.
2.6	**Preferred Embodiment**	Best/preferred version. Additional variations as alternative embodiments — each may become a dependent claim.
2.7	**Filing Type**	Gate: (a) Provisional application first? or (b) Direct Nonprovisional? Set application_type flag.

PHASE 3: PATENT TYPE CLASSIFIER

#	Step	Description / Logic
3.1	**AI Classification**	LLM classifier: (A) Utility — protects function/process, (B) Design — ornamental only, (C) Both. Returns: classification + confidence score + rationale.
3.2	**User Confirmation**	Present AI recommendation with explanation. Allow override. Show comparison table. Warn if wrong type detected.
3.3	**Branch Routing**	IF Design: single-claim workflow, drawings-first. IF Utility: full specification + multi-claim. IF Both: create two linked application records.
3.4	**CPC Classification**	AI suggests CPC codes for examiner routing and prior art targeting. Store: suggested_cpc_codes[] in application record.
3.5	**Technology Domain**	Mechanical / Electrical / Software / Chemical / Biological / Medical Device / Consumer Product. Affects AI templates and claim conventions.

PHASE 4: PRIOR ART SEARCH ENGINE

#	Step	Description / Logic
4.1	**Keyword Extraction**	AI extracts 5–15 key technical terms + synonyms + CPC codes from invention description.
4.2	**USPTO PPUBS Search**	API call: Full-text search (Title, Abstract, Claims) + classification search. Issued patents + published apps 2001+. Top 50 results.
4.3	**Google Patents Search**	Google Patents API — 87M+ patents worldwide. Also run Prior Art Finder across Google Scholar, Google Books, and web. Tag results by source.
4.4	**EPO Espacenet**	EPO Open Patent Services OPS API for international prior art (EP, WO/PCT, DE, JP, CN). Understand global landscape even for US-only filing.
4.5	**AI Relevance Scoring**	LLM evaluates each patent: technical similarity score (0–100), claims overlap, date analysis (is it prior art?), classification overlap. Rank: High/Medium/Low.
4.6	**Novelty Gap Analysis**	AI compares user invention vs. prior art. Output: feature-by-feature matrix, novelty gaps, obviousness risks (103), claim differentiation recommendations.
4.7	**Prior Art Report**	PDF: top 10–20 most relevant patents, why user's invention is novel, suggested claim language to avoid prior art, warning flags if novelty is questionable.
4.8	**IDS Auto-Generation**	Auto-populate Form PTO/SB/08 (Information Disclosure Statement). All material prior art MUST be disclosed. Duty of candor — failure can render patent unenforceable.

PHASE 5: AI DOCUMENT DRAFTING ENGINE

#	Step	Description / Logic
5.1	**System Prompt Build**	Detailed LLM prompt: invention disclosure + patent type + technology domain + prior art context + drafting instructions (USPTO format, no prior art admissions, enabling language). Use claude-opus-4-6 (200K context).
5.2	**Generate Title**	AI generates 3–5 alternatives. User selects or types custom. Validate: 2–7 words, max 500 chars, no articles.
5.3	**Generate Abstract**	Max 150 words enforced. Plain technical language. Key novel features highlighted. Rich text editor with live word counter.
5.4	**Generate Background**	From problem description + prior art context. AI avoids statements admitting prior art weakness that could limit claim scope.
5.5	**Generate Summary**	Parallels broadest independent claim without claims language. Design patents: describes ornamental aspects visible in the drawings.
5.6	**Generate Figure Descriptions**	Per uploaded figure: 'FIG. 1 is a perspective view of the [device]; FIG. 2 is a front elevational view thereof.' Auto-detect view type from image analysis.
5.7	**Generate Detailed Description**	Paragraph-by-paragraph description referencing each numbered element. Structure: Introduction → Component descriptions → Operation → Alternative Embodiments. User can request AI expansion of any section.

PHASE 6: CLAIMS ENGINEERING MODULE

#	Step	Description / Logic

6.1	**Design Auto-Claim**	IF design: auto-generate 'The ornamental design for [TITLE], as shown and described.' Skip to Phase 7.
6.2	**Claim Strategy**	Wizard: independent = broadest protection (fewer elements = broader scope); dependent = fallback + licensable sub-features. Fee alert: >3 independent = $600 surcharge each; >20 total = $200 surcharge each.
6.3	**Independent Claim**	AI drafts Claim 1 (broadest). Format: [Preamble] comprising: [element 1]; [element 2]; wherein [distinguishing feature]. Uses 'comprising' (open). Validates no means-plus-function traps.
6.4	**Dependent Claims**	AI generates 15–19 dependent claims. Strategy: critical features 2–5, embodiments 6–12, specific materials/dimensions 13–19. Drag-and-drop claim reordering.
6.5	**Claims Validator**	Real-time: (1) single sentence, (2) no new matter, (3) proper dependency references, (4) no indefinite language, (5) no improper plural dependent claims. Color-coded status.
6.6	**Prior Art Clearance**	Run each independent claim through patent search. Flag 102 (anticipation) or 103 (obviousness) risks. Suggest amendments to differentiate.
6.7	**Claims Summary**	Count summary with fee calculation. Allow user to consolidate claims to reduce USPTO surcharges.

PHASE 7: DRAWINGS & FIGURES MODULE

#	Step	Description / Logic
7.1	**Drawing Upload**	Drag-and-drop: JPEG, PNG, PDF, SVG, TIF. Multiple files. Auto-assign FIG. numbers. Reorderable. Store in AWS S3.
7.2	**Compliance Check**	AI vision analysis: (a) paper size, (b) margins (top/right 2.5cm, left 3.8cm, bottom 1.0cm), (c) line weight — solid black required, (d) color detection — B&W; required unless petition filed, (e) resolution ≥300 DPI.
7.3	**Reference Numerals**	UI: assign reference numerals to components. Consistency check: same component = same numeral across all figures. Auto-generate component glossary.
7.4	**View Classification**	Classify each figure: Perspective, Front, Back, Left, Right, Top, Bottom, Cross-section, Detail, Exploded, Schematic. Design patents need all standard views — AI flags missing.
7.5	**AI Drawing Generation**	Optional: AI generates B&W; patent-style line drawings from description. User reviews and approves. Recommend professional draftsman verification for final submission.
7.6	**Final Standards Check**	Verify: no photographs (unless petition), reference numeral font ≥0.32cm, lead lines don't cross, each sheet labeled 'Sheet X of Y' with inventor name and figure numbers.
7.7	**Drawing PDF Assembly**	Compile all approved drawings into multi-sheet PDF in USPTO format. Validate final PDF. Export for USPTO submission separately from specification.

PHASE 8: DOCUMENT ASSEMBLY & COMPLIANCE REVIEW

#	Step	Description / Logic
8.1	**Assembly**	Correct USPTO order: (1) ADS [PTO/SB/14], (2) Specification (Title, Cross-Refs, Background, Summary, Figure Descriptions, Detailed Description, Claims, Abstract), (3) Drawings PDF, (4) Oath/Declaration [AIA PTO/AIA/01], (5) IDS [PTO/SB/08], (6) Fee Transmittal [PTO/SB/17]. Output: DOCX + PDF.
8.2	**Completeness Check**	Validate: all sections present and non-empty, abstract ≤150 words, all figures referenced in Detailed Description, all reference numerals match drawings, no duplicate sections, claims supported by specification.
8.3	**Format Validation**	DOCX: margins (top/right 1.5cm, left 2.5cm), Courier or Times 12pt, 1.5/double line spacing, page numbers, 8.5x11 paper. 2025 requirement: nonprovisional utility apps must be filed in DOCX format.
8.4	**Legal Scan**	AI scans: (a) indefinite language ('about', 'approximately', 'substantially' — warn unless defined), (b) new matter in claims vs. spec, (c) 35 USC 112 violations, (d) prior art admissions limiting claim scope.
8.5	**Quality Score**	Score 0–100: specification completeness, claims breadth/defensibility, prior art differentiation, drawing compliance, technical clarity. Report card with actionable improvements.
8.6	**Attorney Review Option**	Optional flat-fee attorney review. Partner attorney annotates in-app. Attorney-client privilege maintained for reviewed portions.
8.7	**Final Review**	Tabbed interface for complete application review. Click any section to jump and edit. 'Mark as Final' locks document. Explicit user confirmation required for each section.

PHASE 9: USPTO FILING INTEGRATION

#	Step	Description / Logic
9.1	**USPTO Account**	OAuth 2.0 with USPTO Patent Center. Verify: eFiler registration, 2FA set up, Customer Number assigned. Support both user's own account and registered agent account.
9.2	**Fee Calculation**	Calculate exact fees: basic filing + search + examination (by entity type), excess claim surcharges, pages-over-100, non-electronic surcharge (avoided by e-filing). Show total before payment. Accept: credit card (Stripe), USPTO Deposit Account.
9.3	**ADS Generation**	Auto-populate Form PTO/SB/14: inventor names/addresses/citizenship, application type, title, priority claims, assignee, attorney info. Output: completed DOCX for upload.
9.4	**DOCX Prep**	Validate specification DOCX: correct styles, no tracked changes, no comments, no embedded images in spec body. Convert if needed. New 2025 requirement: nonprovisional utility apps in DOCX.
9.5	**Pre-Submission Checklist**	Final automated checklist: ADS complete, Specification DOCX, Drawings PDF (separate), Declaration signed, IDS submitted, Fee authorized, Receipt email configured.
9.6	**Patent Center Submit**	Submit via USPTO Patent Center API: upload all documents, submit fee, confirm. Capture: application number, filing date, receipt number. Send email confirmation.
9.7	**Filing Receipt**	Parse receipt: extract application number (e.g., 29/612,103), official filing date. Store receipt PDF. Calculate all downstream deadlines: 12-month provisional deadline, PCT deadline, expected exam date.

PHASE 10: APPLICATION TRACKING & POST-FILING MANAGEMENT

#	Step	Description / Logic
10.1	**Status Monitor**	Poll USPTO Patent Center API daily. Alert via email + in-app: new correspondence, office action, allowance/rejection, deadlines (30/60/90 day warnings).
10.2	**Office Action Analysis**	AI reads rejections: 102 (anticipation), 103 (obviousness), 112 (specification issues), 101 (patent eligibility for software/biotech). Plain-English summary of examiner's objections.
10.3	**Response Drafting**	For 102/103: suggest claim amendments + written arguments distinguishing prior art. For 112: spec amendments. For drawing objections: correction guide. Track deadline + extension costs.
10.4	**RCE & Continuations**	Alert when RCE is available. Track continuation/divisional/CIP opportunities. Remind of PCT deadline (12 months from priority) for international protection.
10.5	**Patent Grant**	On allowance: calculate issue fee, generate payment reminder. After grant: display patent number, maintenance fee calendar (utility: 3.5/7.5/11.5 years), generate certificate PDF. Note: design patents have NO maintenance fees.
10.6	**Portfolio Dashboard**	All applications + status, key deadlines calendar, fee payment schedule, prior art portfolio, assignee/licensing tracking. Export portfolio summary report.

5. APPLICATION ARCHITECTURE

5.1 System Layers

Layer	Component	Recommended Technology
Frontend	User Interface / Wizard	React.js + TypeScript + Next.js, TailwindCSS, Shadcn/UI
Frontend	Rich Text Editor	TipTap (ProseMirror-based) — custom patent document schemas
Frontend	Drawing Annotation	Konva.js or Fabric.js for figure reference numeral overlays
API Gateway	REST API	FastAPI (Python 3.12) — async, auto-docs, AI integration
AI / LLM	Patent Drafting	Anthropic Claude API (claude-opus-4-6, 200K context window)
AI / LLM	Drawing Analysis	OpenAI Vision API (GPT-4V) for compliance checking
AI / LLM	Classification	HuggingFace patent classifier + CPC code prediction model
Search Layer	Patent Search Aggregator	Custom service: USPTO PPUBS + Google Patents + EPO OPS APIs
Search Layer	Semantic Patent Search	text-embedding-3-large + Pinecone vector database
Document Engine	DOCX Generation	python-docx + DOCX templates per patent type
Document Engine	PDF Output	ReportLab (Python) for user-facing formatted documents
Primary Database	Relational Store	PostgreSQL 16 + pgvector (relational + vector similarity)
Document Store	Draft Versioning	MongoDB Atlas (flexible schema for evolving patent drafts)
File Storage	Drawings, PDFs, DOCX	AWS S3 — versioning, encryption, per-user isolation
Authentication	User Auth + MFA	Auth0 — OAuth 2.0, FIDO2/WebAuthn, USPTO OAuth linking
Payments	Fee Collection	Stripe — PCI-compliant, subscriptions + one-time USPTO fees
E-Signature	Declaration Forms	DocuSign — UETA/ESIGN compliant, USPTO-acceptable
Cloud	Hosting	AWS ECS/Fargate (backend) + Vercel (frontend)
Monitoring	Observability	Datadog — metrics, logs, traces, USPTO API latency alerts

5.2 Data Flow

```
1. User Input (Invention Description + Drawings)
2. Intake Module → PostgreSQL (applications table)
3. AI Classifier → Determine patent_type, technology_domain, cpc_codes
4. Search Service → USPTO PPUBS + Google Patents + EPO APIs → prior_art table
5. LLM Drafting Engine (Claude API) → Generate spec sections → documents table
6. Claims Engine → Validate → claims table
7. Drawing Processor → S3 → Compliance check → figures table
8. Assembly Engine → Generate DOCX + PDF → Compliance audit
9. USPTO Patent Center API → Filing receipt → Confirmation stored
```

```
10. Daily Status Monitor → Alert system → Response drafting assistant
```

6. DATABASE SCHEMA (PostgreSQL)

Table: users

```
id UUID PRIMARY KEY
email VARCHAR(255) UNIQUE NOT NULL
full_name VARCHAR(255)
entity_type VARCHAR(10) -- 'large','small','micro'
is_attorney BOOLEAN DEFAULT false
attorney_reg_number VARCHAR(50)
citizenship VARCHAR(100)
address JSONB
kyc_verified BOOLEAN DEFAULT false
```

Table: applications

```
id UUID PRIMARY KEY
user_id UUID REFERENCES users(id)
title VARCHAR(500)
patent_type VARCHAR(20) -- 'utility','design','plant','provisional'
technology_domain VARCHAR(100)
suggested_cpc_codes TEXT[]
status VARCHAR(20) -- 'draft','reviewing','filed','pending','allowed','granted','abandoned'
application_number VARCHAR(20) -- USPTO assigned after filing
filing_date DATE
entity_type VARCHAR(10)
total_claims INT DEFAULT 0
independent_claims INT DEFAULT 0
created_at TIMESTAMP
updated_at TIMESTAMP
```

Table: specification_sections

```
id UUID PRIMARY KEY
application_id UUID REFERENCES applications(id)
section_type VARCHAR(50) -- 'title','background','summary','figure_descriptions',
-- 'detailed_description','claims','abstract','cross_references'
content TEXT
ai_generated_at TIMESTAMP
user_approved_at TIMESTAMP
word_count INT
version INT DEFAULT 1
```

Table: claims

```
id UUID PRIMARY KEY
application_id UUID REFERENCES applications(id)
claim_number INT NOT NULL
claim_type VARCHAR(15) -- 'independent','dependent'
depends_on INT -- references claim_number for dependent claims
claim_text TEXT NOT NULL
validation_status VARCHAR(10) -- 'valid','warning','error'
validation_notes TEXT
```

Table: prior_art

```
id UUID PRIMARY KEY
application_id UUID REFERENCES applications(id)
```

```
patent_number VARCHAR(30)
title VARCHAR(500)
inventors TEXT
assignee VARCHAR(255)
filing_date DATE
grant_date DATE
source VARCHAR(20) -- 'uspto','google_patents','espacenet','other'
relevance_score INT
relevance_notes TEXT
cited_in_ids BOOLEAN DEFAULT false
```

Table: figures

```
id UUID PRIMARY KEY
application_id UUID REFERENCES applications(id)
figure_number INT
view_type VARCHAR(20) -- 'perspective','front','back','left','right','top','bottom'
s3_url VARCHAR(500)
file_format VARCHAR(10)
resolution_dpi INT
compliance_status VARCHAR(10) -- 'pass','warning','fail'
caption TEXT
```

Table: office_actions

```
id UUID PRIMARY KEY
application_id UUID REFERENCES applications(id)
action_date DATE
response_deadline DATE
rejection_types TEXT[] -- ['102','103','112','101']
ai_analysis TEXT
ai_draft_response TEXT
response_filed_at TIMESTAMP
```

7. API INTEGRATIONS

API / Service	Purpose	Auth	Key Notes
USPTO Patent Public Search	Prior art — all US patents 1790–present	Public API key	Full-text + classification search
USPTO Patent Center API	E-filing, status, document retrieval	OAuth 2.0 + USPTO account	File apps, receipts, exam status
Google Patents API	International — 87M+ patents	Google Cloud key	Patent metadata + Prior Art Finder
EPO Open Patent Services (OPS)	European + PCT international prior art	OAuth 2.0 client credentials	EP, WO, DE, JP, CN patent data
Anthropic Claude API	AI drafting, classification, OA analysis	API key (claude-opus-4-6)	200K context for full patent docs
OpenAI Vision API	Drawing compliance + AI figure generation	API key	GPT-4V for images; DALL-E 3 for figures
Stripe	Payment — subscription + USPTO fees	Stripe API key + webhook	PaymentIntent, Subscription, Customer
Auth0	User auth, MFA, USPTO account linking	OAuth 2.0	Pools, tokens, MFA, FIDO2
AWS S3	Drawings, PDFs, DOCX storage	IAM role	Pre-signed URLs, versioning, isolation
DocuSign	E-signature on Oath/Declaration	OAuth 2.0	Envelope routing, completion webhooks
KYC (Persona/Jumio)	Inventor identity verification	API key	Doc verify, liveness, ID scan
SendGrid	Transactional email — deadlines, alerts	API key	Templates, delivery tracking

8. FEE CALCULATOR MODULE

Fees per USPTO schedule effective January 19, 2025 (last revised May 1, 2026). Store fees in a fees_config database table — update quarterly from the official USPTO fee schedule PDF.

Fee Type	Large Entity	Small Entity	Micro Entity
Provisional — Basic Filing	$320	$128	$64
Utility — Filing	$350	$140	$70
Utility — Search	$700	$280	$140
Utility — Examination	$800	$320	$160
Utility — Excess independent claim (per claim over 3)	$600	$240	$120
Utility — Excess total claims (per claim over 20)	$200	$80	$40
Utility — Non-electronic filing surcharge (avoided by e-filing)	$400	$200	$200
Design — Filing + Search + Examination (total)	$1,300	$520	$260
Utility — Issue Fee (upon grant)	$1,200	$480	$240
Design — Issue Fee (upon grant)	$900	$360	$180
Utility — Maintenance at 3.5 years	$3,200	$1,600	$800
Utility — Maintenance at 7.5 years	$7,200	$3,600	$1,800
Utility — Maintenance at 11.5 years	$14,800	$7,400	$3,700
IDS — after 3 months from filing	$360	$144	$72

Fee Calculator Formula: total_filing_fee = filing_fee + search_fee + exam_fee + max(0, indep_claims - 3) * excess_indep_fee + max(0, total_claims - 20) * excess_claim_fee + (non_elec_surcharge if paper_filing else 0). Multiplier: Large=1.0x | Small=0.4x | Micro=0.2x.

9. SECURITY & COMPLIANCE

Data Encryption: All patent content encrypted at rest (AES-256) and in transit (TLS 1.3). Patent applications are extremely sensitive, commercially valuable IP.

Data Isolation: Row-level security in PostgreSQL. Separate S3 prefixes per user. Users must NEVER be able to access each other's patent content. Multi-tenant security audit required before launch.

Unauthorized Practice of Law: Frame as 'document preparation assistance' — not legal advice. Prominent disclaimers on every page. Attorney review option for high-value applications.

AI Audit Trail: All AI-generated content logged: timestamp, model version, prompt hash, token count. Users can see AI-generated vs. user-written text.

SOC 2 Type II: Target SOC 2 certification for enterprise. Access controls, audit logging, change management, incident response.

GDPR / CCPA: Minimum PII collection. Data deletion on request. Cookie consent. EU data residency option planned.

Filing Authentication: Step-up authentication before any USPTO filing action: FIDO2/WebAuthn required. No application filed without explicit 2FA confirmation.

E-Signature Standards: Oath/Declaration via DocuSign — UETA/ESIGN compliant. Must meet USPTO AIA e-signature standards (37 CFR 1.4(d)(3)).

Backup & DR: Daily automated backups. Point-in-time recovery. RPO: 1 hour, RTO: 4 hours. Cross-region replication. Quarterly restore testing.

10. RECOMMENDED TECHNOLOGY STACK

Category	Technology	Justification
Frontend Framework	React.js + TypeScript + Next.js	SSR/SEO, type safety, large ecosystem
UI Components	Shadcn/UI + TailwindCSS	Accessible, customizable, patent form components
Rich Text Editor	TipTap (ProseMirror)	Custom patent schemas, track changes, collaborative editing
Backend API	FastAPI (Python 3.12)	Async, AI integration, auto API docs
AI / LLM	Anthropic Claude API (claude-opus-4-6)	Best for long-form technical writing, 200K context
Document — DOCX	python-docx	USPTO-compliant DOCX with styles and templates
Document — PDF	ReportLab (Python)	Professional patent-formatted PDF output
Primary Database	PostgreSQL 16 + pgvector	Relational data + vector similarity in one DB
Document Store	MongoDB Atlas	Flexible schema for patent draft versions
File Storage	AWS S3	Reliable, cheap, versioning, encryption at rest
Vector Search	Pinecone + text-embedding-3-large	Semantic patent similarity for prior art search
Authentication	Auth0	Enterprise MFA, FIDO2, USPTO OAuth linking
Payments	Stripe	PCI-compliant, subscriptions + one-time USPTO fees
E-Signature	DocuSign API	UETA/ESIGN compliant, USPTO-acceptable, audit trail
Cloud Platform	AWS (primary)	SOC 2, HIPAA-eligible, global regions
CI/CD	GitHub Actions + AWS CodeDeploy	Automated tests, security scan, blue-green deploy
Monitoring	Datadog	Full-stack observability, USPTO API latency, alerts

11. DEVELOPMENT ROADMAP

Phased approach: start with design patents (simpler, single claim, drawings-centric) before expanding to full utility patent support and international filings.

MVP — Months 1–3: Design Patent Focus

- User registration, KYC, entity type determination
- Invention disclosure intake — design-specific workflow
- Auto-generated single ornamental claim
- Drawing upload + 37 CFR § 1.84 compliance checking
- Prior art search — USPTO PPUBS integration
- Document assembly → PDF export
- Guided manual USPTO filing instructions (not yet automated)
- Basic fee calculator with entity type selection

Version 1.0 — Months 4–6: Full Utility Patent Support

- Full utility patent specification drafting (all 9 sections)
- Claims engineering module (independent + dependent, real-time validation)
- AI-powered prior art analysis and novelty gap report
- IDS auto-generation (Form PTO/SB/08)
- Direct USPTO Patent Center filing API integration
- Stripe payment for USPTO fees
- Application status tracking dashboard
- DOCX format compliance (2025 USPTO requirement)

Version 1.5 — Months 7–9: AI Enhancement + Office Actions

- Office Action AI analysis and response drafting assistant
- Attorney review marketplace integration (flat-fee)
- Collaborative editing for multiple inventors
- Provisional → Nonprovisional conversion workflow
- DocuSign e-signature for oath/declaration
- Multi-patent portfolio management dashboard
- Advanced claims strategy AI with claim tree visualization

Version 2.0 — Months 10–12: International Expansion

- PCT (Patent Cooperation Treaty) international filing workflow
- European Patent Office (EPO) direct filing integration
- Multi-language support (Spanish, Mandarin, German, Hebrew)
- Enterprise API for law firms and corporate IP departments
- AI drawing generation from written description (DALL-E/specialized)
- Patent portfolio analytics and estimated valuation

- White-label solution for corporate IP departments

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